

# Lecture 12 - Autism

The Social Brain: Critical Perspectives on Science, Society and Neurodiversity

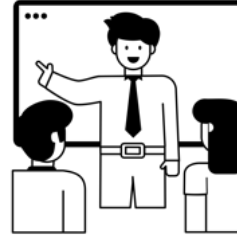
Richard Ramsey



# Today

## Part 1

- Autism



## Part 2

- Read articles and discuss



# Overview

- Background
- What factors cause autism?
  - Environmental
  - Genetic
  - Cognitive
  - Neurobiological
- Many autisms?



# Background



# What is autism?

- Lorna Wing's triad of impairments:
  - difficulties in social interaction
  - difficulties in communication
  - lack of imagination / repetitive interests
- Heterogeneous – no two people are the same.
- There is a spectrum of different diagnoses.



# Diagnosis

- DSM criteria (Diagnostic and Statistical Manual of Mental Disorders, APA)
- Clinical observation
- Checklists
  - CHAT (checklist for autism in toddlers)
  - CARS (childhood autism rating scale)
  - DISCO (Diagnostic Interview for Social and Communication Disorders)
- ADOS (Autism Diagnostic Observation Schedule)
- ADI (Autism Diagnostic Interview)



# How many children have autism?

- 3–4% of children and adolescents in the US
- 1% of the global population
- 0.7–1.5% in Switzerland
- Male-to-female ratio of approximately 3–4:1.



# What factors cause autism?



# Possible factors

- environmental
  - ~~vaccines~~
  - other environment?
- genetic
- cognitive
- neurobiological



# Vaccines

- **HELL NO!**
- There is a vast amount of high quality evidence that shows **NO LINK** between vaccines and autism.
- For those who want to read more see [here](#).



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(DeStefano & Shimabukuro 2019; Gahle et al. 2022)

# Other environmental factors

- Can a lack of a normal social environment cause autism?
  - “Refrigerator mother” - Bruno Bettelheim (1950’s) blamed “lack of maternal warmth” for autism in a child.
  - Theory is now totally discredited.
- What about sensory deprivation?
  - Blind and deaf children show autistic-like symptoms (Brown et al., 1997; Wolfe et al., 2002).
  - Neither blind nor deaf show all the symptoms of autism.
- Or extreme social deprivation?
  - Romanian orphanage studies (Rutter et al., 2007).
  - 10% had some autism symptoms at age 6, none had full symptoms, and many improved by age 12



# Genetic

- Bailey et al, 1995
  - 60% of MZ twin pairs were concordant for autism
  - no DZ twin pairs were concordant
  - shows high heritability (but not 100%)
- Ronald et al, 2005
  - studied over 3000 twin pairs in TEDS
  - find social behaviour is 62-76% heritable

Many genes have been implicated, none conclusively.



# Cognitive

Two different cognitive accounts:

- Theory of mind
- Weak central coherence

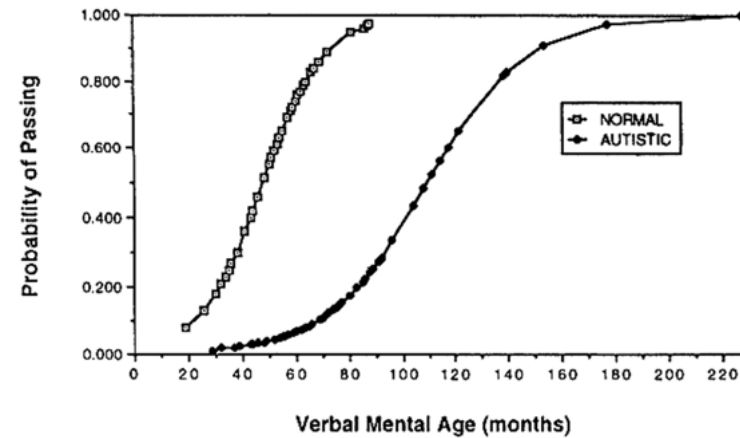
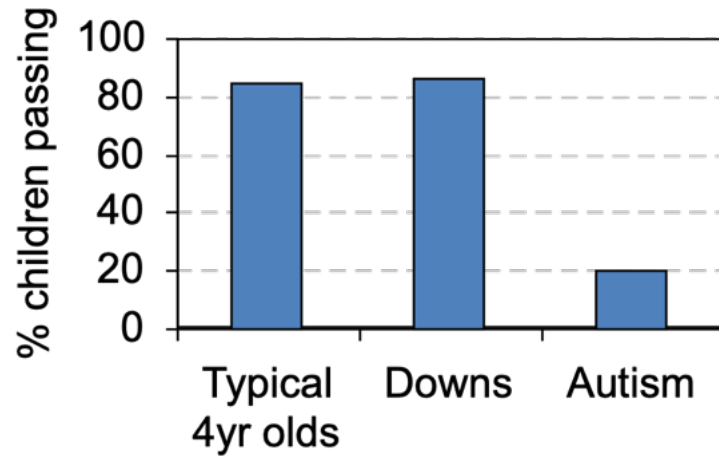


# Theory of mind

- The idea that people with autism have difficulty understanding other people's mental states (beliefs, desires etc.).
- Proponents include Frith, Leslie, Happe, Baron-Cohen



# False beliefs



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(Baron-Cohen et al 1985, Happe 1995)

# Animated shapes

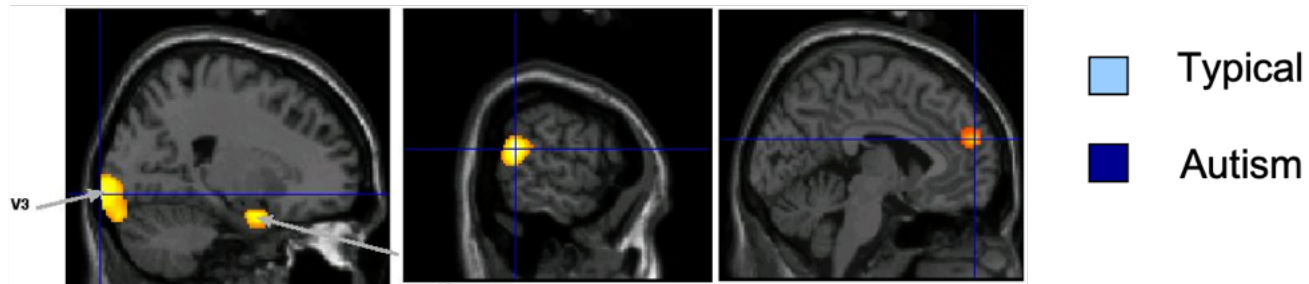
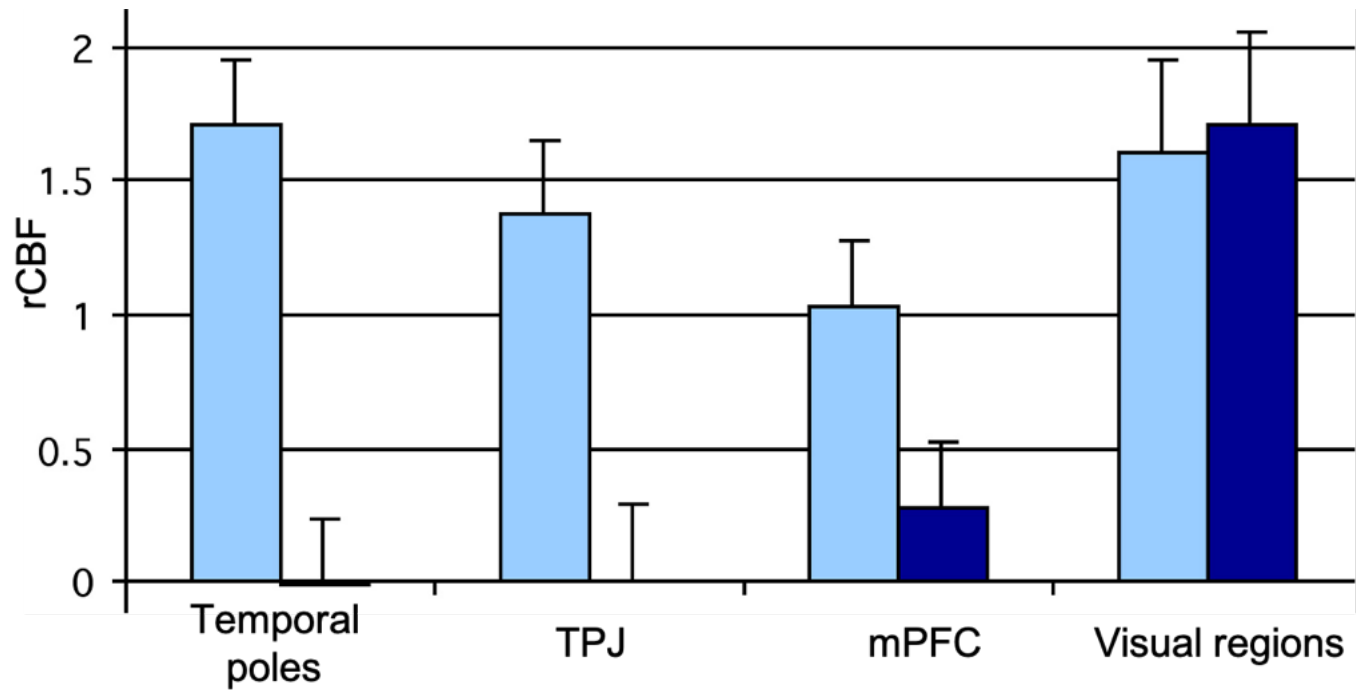
Heider and Simmel (1944) animation



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(Heider & Simmel 1944)

# Animated shapes, autism and fMRI

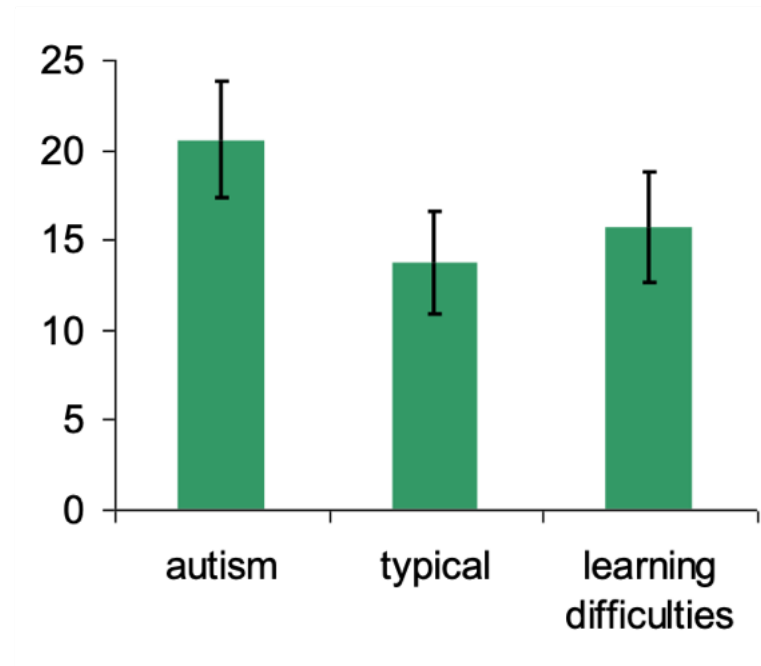
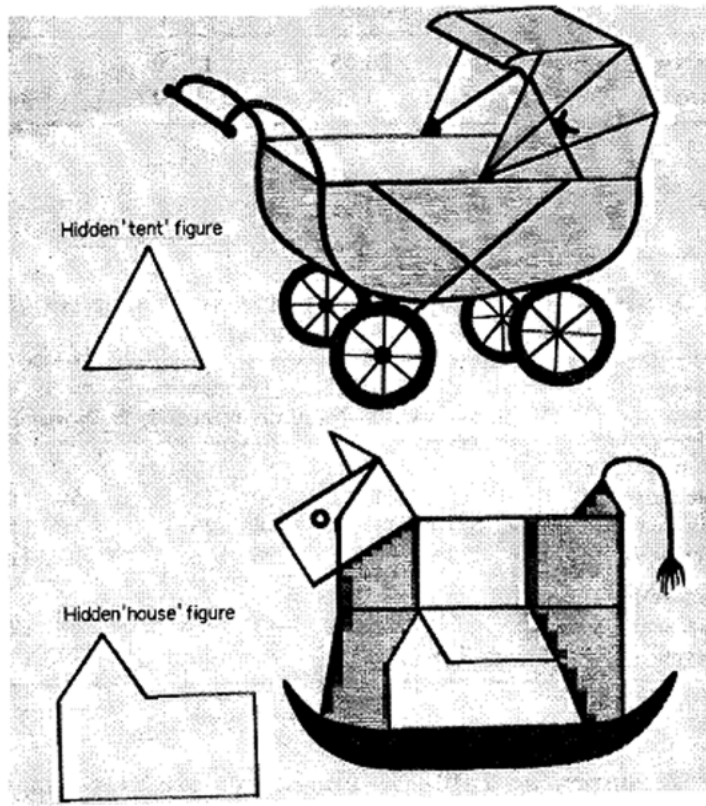


# Weak central coherence

- The idea that people with autism have superior processing of detail and less superior processing of whole contexts.
- It can account for many non-social features and strengths.
- Do social tasks require more 'central coherence' than non-social tasks?

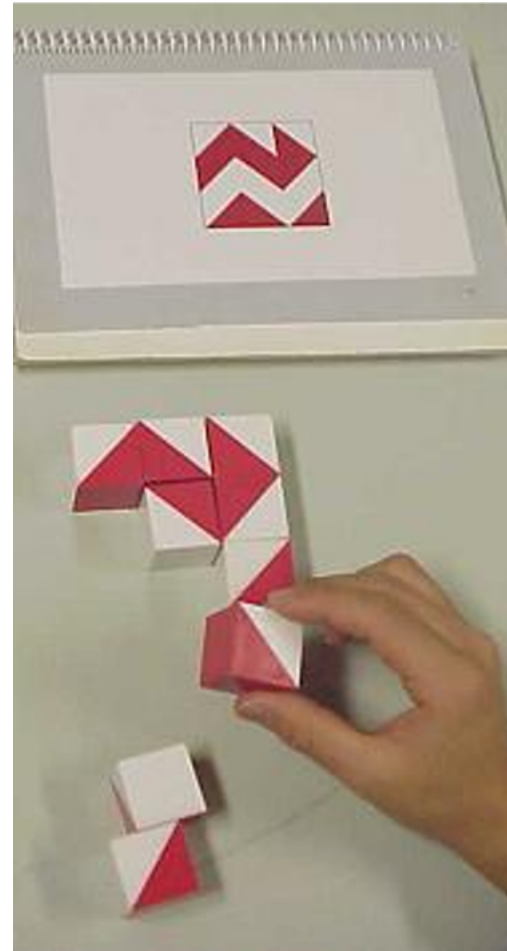


# The emebbeded figures task



# The block design task

- Part of WISC IQ test
- Children with autism often perform very well on this task



(Shah & Frith 1992)

# Cognitive summary

- ToM
  - Lack of ToM in autism could cause breakdown in social interaction and communication
  - But it can't explain savant skills or repetitive behaviours
- Weak central coherence
  - Weak central coherence could maybe account for many non-social features and strengths.
  - But do social tasks require more 'central coherence' than non-social tasks?



# Neurobiological

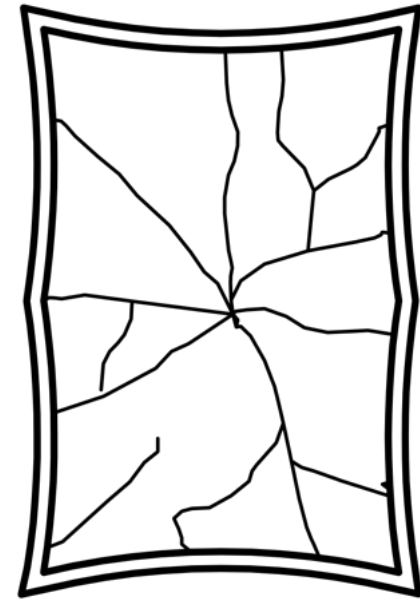
Two different neurobiological accounts:

- Broken mirror theory
- Underconnectivity

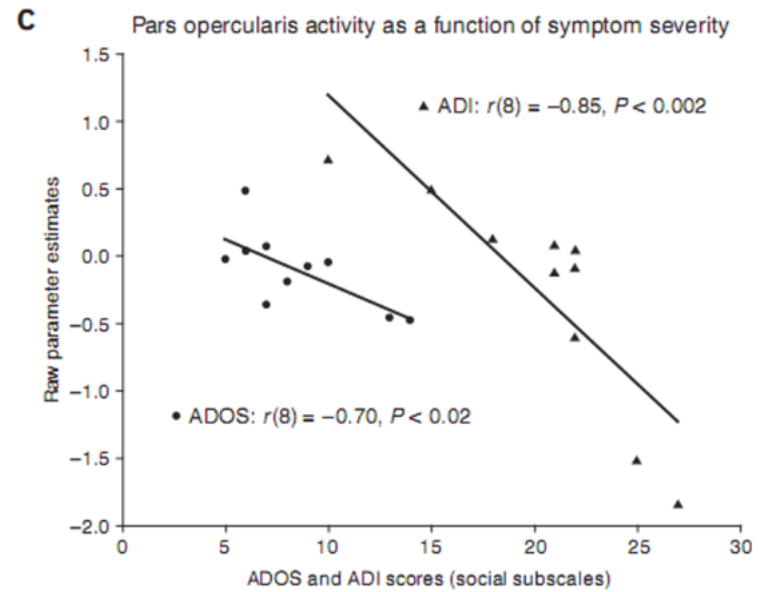
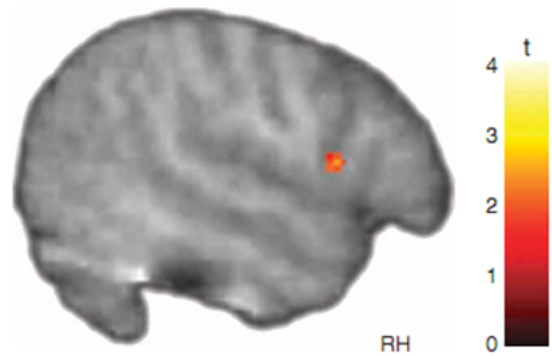


# Broken mirrors

- The idea that the ‘mirror neuron system’ is dysfunctional in autism
- Proponents include: Dapretto & Iacoboni, Oberman & Ramachandran, Justin Williams
- Evidence from: imitation behaviour, fMRI, EEG



# fMRI evidence



(Dapretto et al 2006)

# Broken mirror criticisms

- Evidence for the theory is weak ([Southgate & Hamilton A. F., 2008](#)).
  - emotion and language tasks used, which are not the like the tasks that activate the mirror neurons in monkeys
  - types of imitation remain intact
- Empirical evidence that EEG and mu rhythmns can serve as an index of mirror neuron system has been heavily questioned ([Hobson & Bishop, 2016, 2017](#)).

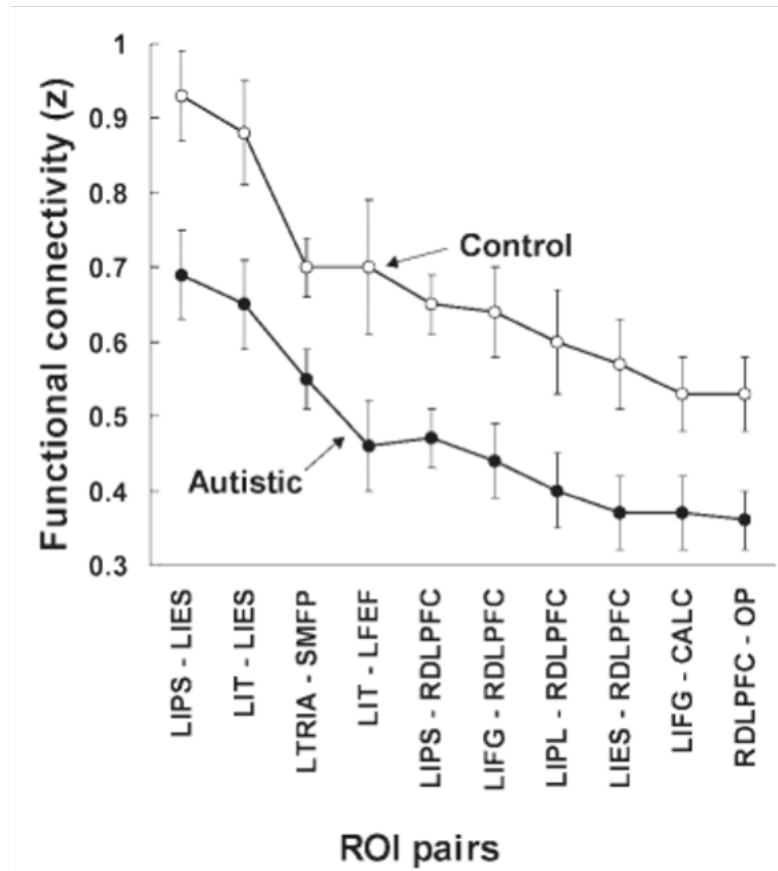


# Underconnectivity

- The idea that the cortex is not connected together correctly in autism
  - Underconnectivity could cause lots of information processing problems
- Proponents include Minshew, Just, Courschesne
- Evidence comes from fMRI studies



# fMRI evidence



A Autism group



B Control group



# Neurobiological summary

- Broken mirror theory claims to explain many social features of autism, but it does not explain non-social features.
- And ...
  - We still don't know if the MNS contributes to Theory of Mind or language.
  - Mostly indirect measures of MNS function.
  - Tasks are emotional or meaningless actions – not like the tasks that activate the monkey MNS.
- Underconnectivity may explain weak central coherence
- But ...
  - it may not explain social / communication problems (unless social information processing requires more 'connections')



# Many autisms?

- No single factor gives a complete explanation
- It is hard to explain both social and non-social symptoms

*Time to give up on a single explanation for autism (Happé et al., 2006)*

- Should study each part of the triad separately
- No model can yet account for all symptoms
- Different genetic and cognitive causes for each



# Take a break



# Part 2 - Read and discuss



# Discussion material

- break into small groups (~ 5 per group)
- discuss aspects of the lecture
- discuss aspects of the journal article: <https://www.nature.com/articles/nn1770>

And some articles in the popular press:

Vaccines do not cause autism

So, what is going on in the United States?



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